

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

We learn more about ISRO's Mangalyaan mission. Then, we take a look at how Gallium Nitride could save India's semiconductor dream. Finally, we explore some of India's natural wonders.



No :P for the T-Rex

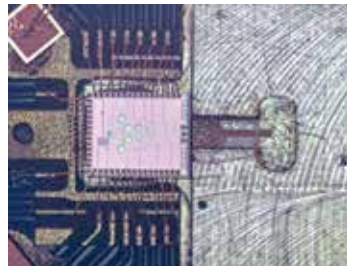
Comparing the fossils of dinosaurs to skulls of birds and crocodiles has showed that the dinosaurs and pterosaurs could not stick out their tongues like lizards do. The tongues were probably attached to the base of the skulls, similar to the tongues of alligators. This means that all the depictions of dinosaurs with their tongues sticking out of their faces are wrong.

<https://digit.in/dinotong>

WHAT'S NEW

New molecular clock to improve smartphone navigation accuracy

There are times when a signal from navigation satellites are not available – say, if the device is surrounded by buildings that reflect the signals, or is going through a tunnel, or is underwater. In such situations, a crystal-oscillator clock on the device is used in combination with an accelerometer to provide the location. The more accurate this clock is, the more accurate the location tracking. Researchers from MIT have developed a molecular clock on a chip, that can be



used in smartphones to increase the accuracy of navigation. A GHz signal produced by a clock transmitter chip excites the rotation of carbonyl sulfide molecules in a tiny gas cell. The period of rotation of the molecules is a constant, and can be used to

track units of time, such as seconds. The resulting clock is about a 10,000 times more stable than the clocks currently used on smartphones, and is comparable to the accuracy of atomic clocks.

<https://digit.in/mitmc>

105 million year old snake found preserved in amber

A team of international researchers have found the oldest baby snake preserved in amber. The contents of the amber also reveal the environment in which the snake lived – a forest area with plenty of insects. The find illuminates the early history of snakes, particularly their spread through an ancient continent known as Gondwana. There are only

20 or so fossilised snakes that are 100 million years old. The fossilised snake is related to ancient snake species in Argentina, Africa, India and Australia. The team used CT scans to study the snake. Their studies helped advance the knowledge of the development



and embryology of ancient snakes, including the formation of the vertebrae and the notochord. The research has been published in a paper titled "A

Mid-Cretaceous Embryonic-to-Neonate Snake in Amber From Myanmar" in Science Advances. <https://digit.in/babysnek>



Origin of Neutrinos found

Researchers from Drexel have successfully identified the origin of Neutrinos: Active galactic nuclei that fire high energy ionized matter, often directly towards the Earth, known as blazars. The oval shaped galaxies are believed to have supermassive black holes at their centres.

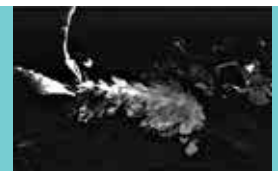
<https://digit.in/ntrno>



Water locked away in the mantle

This diamond, formed deep within the Earth, contains embedded within it, microscopic flecks of water bearing minerals. The finding indicates that between 410 and 660 kilometres below the surface of the planet, there is a layer of water rich minerals.

<https://digit.in/waterworld>



Dragonflies inspire heart valves

Dragonfly larvae inhale water, extract the oxygen, and then release the water through a valve that is very similar to human heart valves. Inspired by these larvae, researchers plan to design a heart valve that can direct jets of blood in specific directions.

<https://digit.in/dflyheart>